

# SIMATS ENGINEERING

## ASSESSMENT TOOL II – DATA ANALYSIS PROJECT

**Course:** Data Handling and Visualization

**Code:** DSA0601

**CO:** CO3

**Duration:** 120 min

**Total Marks:** 100

**Reg No:** 192424322

**NAME:** M.Arshini

### COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

**Activity Tool :** Data Analysis Project

**Weightage:** 20%

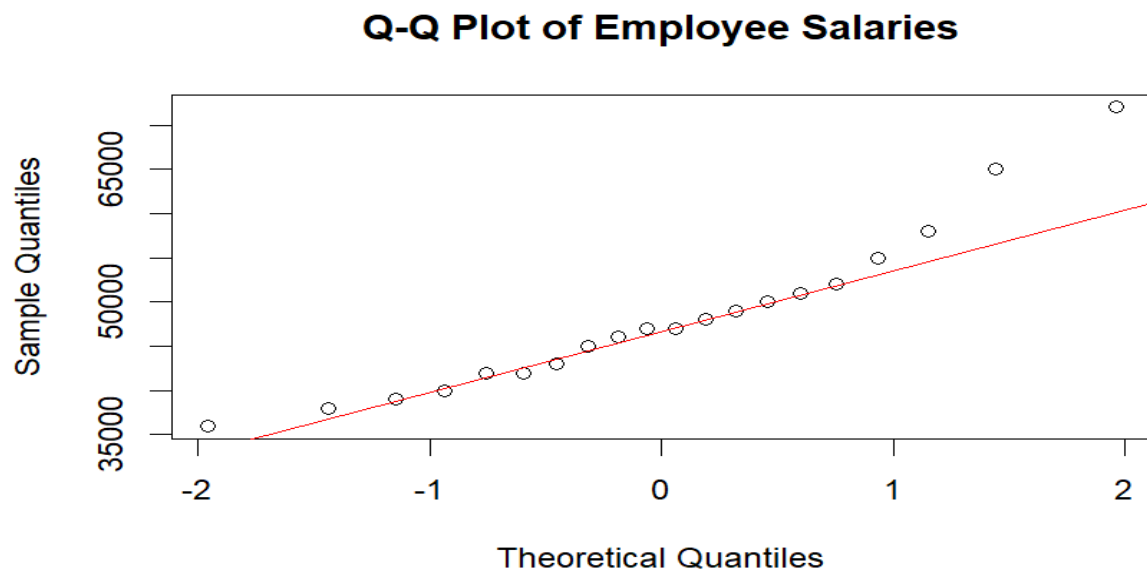
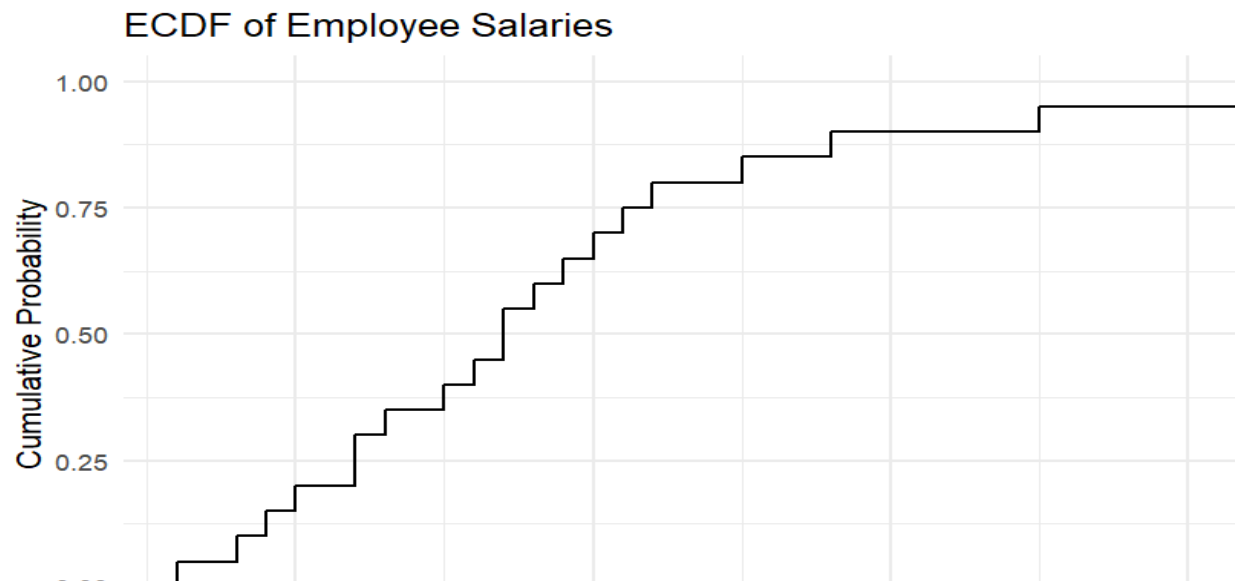
### Code of Conduct:

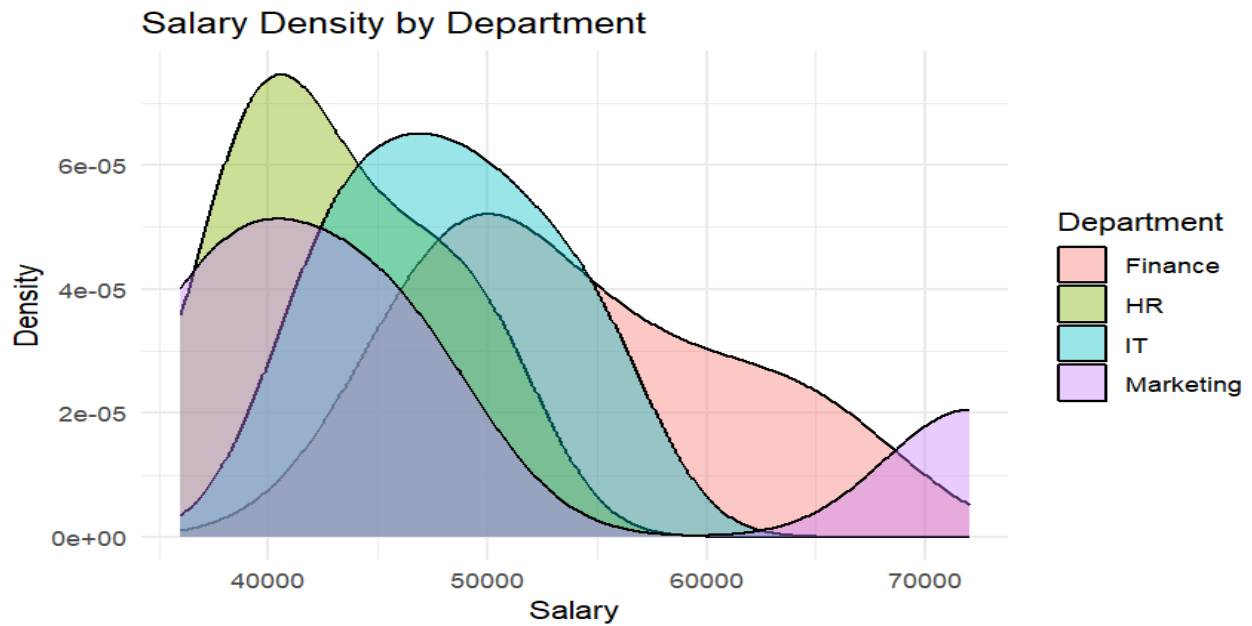
I **M.Arshini(192424322)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Nithya Sree

| Criteria | Problem Understanding & Data Preparation<br>25% | Data Analysis & Interpretation<br>30% | Application of Analytical Techniques<br>20% | Analysis of Results<br>15% | Organization & Documentation<br>10% | Presentation & Clarity<br>5% | Total |
|----------|-------------------------------------------------|---------------------------------------|---------------------------------------------|----------------------------|-------------------------------------|------------------------------|-------|
| Q1       |                                                 |                                       |                                             |                            |                                     |                              |       |
| Q2       |                                                 |                                       |                                             |                            |                                     |                              |       |
| Q3       |                                                 |                                       |                                             |                            |                                     |                              |       |
| Q4       |                                                 |                                       |                                             |                            |                                     |                              |       |
| Q5       |                                                 |                                       |                                             |                            |                                     |                              |       |

### QUESTION-1 ECDF, Q-Q Plot and Department Salary Distribution





CODE:

```
library(ggplot2)
library(dplyr)
# Create dataset
salary <- data.frame(
  Employee_ID = paste0("E", sprintf("%03d", 1:20)),
  Department = c(
    rep("IT", 5),
    rep("HR", 5),
    rep("Finance", 5),
```

```

rep("Marketing", 5) ),
Salary = c(
  42000, 45000, 48000, 51000, 55000,
  38000, 40000, 42000, 46000, 50000,
  47000, 49000, 52000, 58000, 65000,
  36000, 39000, 43000, 47000, 72000))
print(salary)
ggplot(salary, aes(x = Salary)) +
  stat_ecdf() +
  labs(
    title = "ECDF of Employee Salaries",
    x = "Salary",
    y = "Cumulative Probability" ) +
  theme_minimal()
qqnorm(
  salary$Salary,
  main = "Q-Q Plot of Employee Salaries")
qqline(
  salary$Salary,
  col = "red")
ggplot(salary, aes(x = Department, y = Salary)) +
  geom_boxplot() +
  labs(
    title = "Salary Distribution by Department",
    x = "Department",
    y = "Salary" ) +
  theme_minimal()
ggplot(salary, aes(x = Salary, fill = Department)) +
  geom_density(alpha = 0.4) +
  labs(
    title = "Salary Density by Department",
    x = "Salary",
    y = "Density" ) +
  theme_minimal()

```

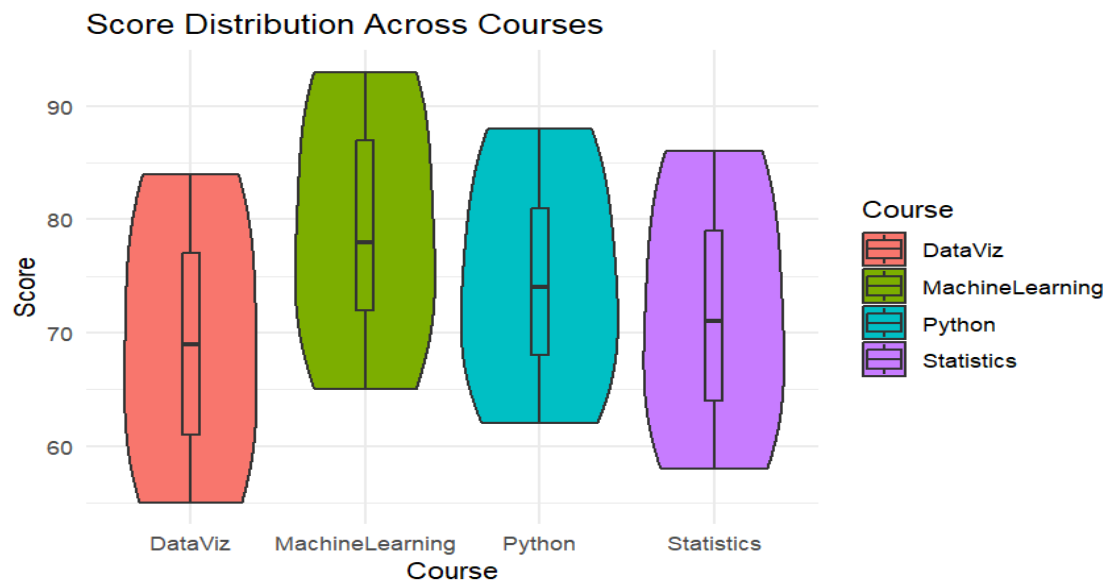
## Interpretation

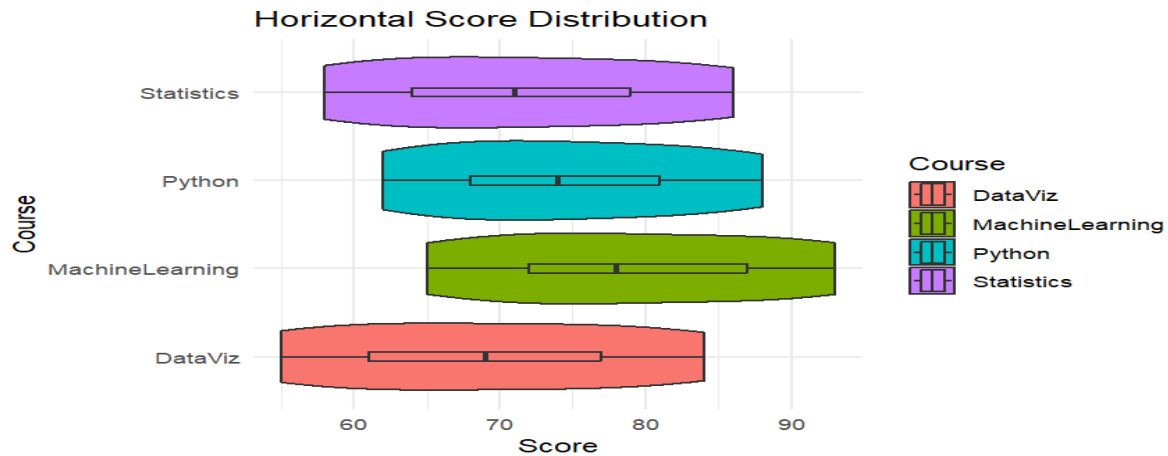
- The **ECDF** shows the cumulative proportion of employees earning below each salary level.
- The **Q-Q plot** checks whether salaries follow a normal distribution.
- Points that move away from the straight line indicate deviation from normality.
- **Marketing** has an unusually high salary of **72,000**, which may contribute to right skewness.
- **Finance** also has relatively high salaries, especially 65,000.
- Marketing has the largest salary range.

## Important values

| Department | Minimum | Maximum |
|------------|---------|---------|
| IT         | 42,000  | 55,000  |
| HR         | 38,000  | 50,000  |
| Finance    | 47,000  | 65,000  |
| Marketing  | 36,000  | 72,000  |

## QUESTION 2 – Bean/Violin Plots





CODE:

```
library(ggplot2)
# Create dataset
student <- data.frame(
  Student_ID = paste0("S", sprintf("%03d", 1:20)),
  Course = c(
    rep("Python", 5),
    rep("Statistics", 5),
    rep("DataViz", 5),
    rep("MachineLearning", 5)),
  Study_Hours = c(
    5, 7, 9, 11, 14,
    4, 6, 8, 10, 13,
    3, 5, 7, 9, 12,
    6, 8, 10, 13, 16 ),
  Score = c(
    62, 68, 74, 81, 88,
    58, 64, 71, 79, 86,
    55, 61, 69, 77, 84,
    65, 72, 78, 87, 93 ))
# View data
print(student)
ggplot(student, aes(x = Course, y = Score, fill = Course)) +
  geom_violin() +
  geom_boxplot(width = 0.1) +
  labs(
    title = "Score Distribution Across Courses",
    x = "Course",
    y = "Score"
```

```

) +
theme_minimal()
ggplot(student, aes(x = Score, y = Course, fill = Course)) +
  geom_violin() +
  geom_boxplot(width = 0.1) +
  labs(
    title = "Horizontal Score Distribution",
    x = "Score",
    y = "Course"
  ) +
  theme_minimal()

```

## Interpretation

- The violin plot shows the **shape and spread** of scores.
- **MachineLearning** has the highest scores overall.
- MachineLearning scores range from **65 to 93**.
- **DataViz** has the lowest scores overall.
- Python has a relatively high and consistent performance.
- The courses have similar score ranges, but their centers differ.

## Highest-performing course

### MachineLearning

Average scores:

Python = 74.6

Statistics = 71.6

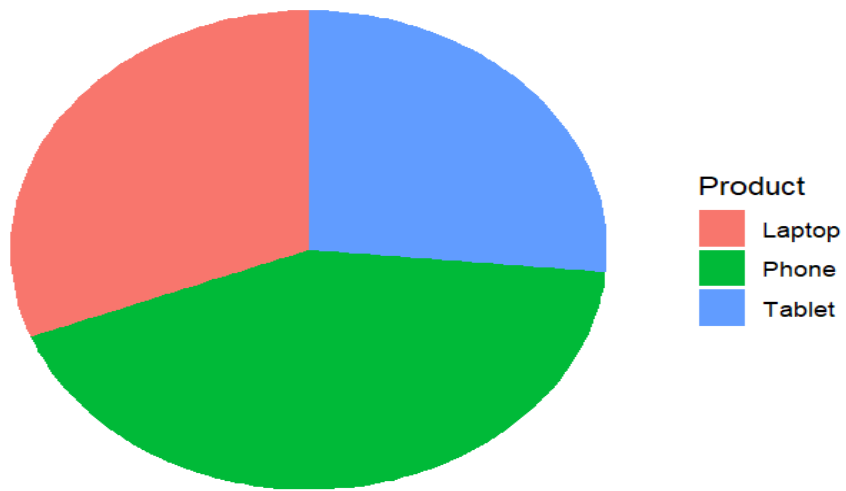
DataViz = 69.2

MachineLearning = 79.0

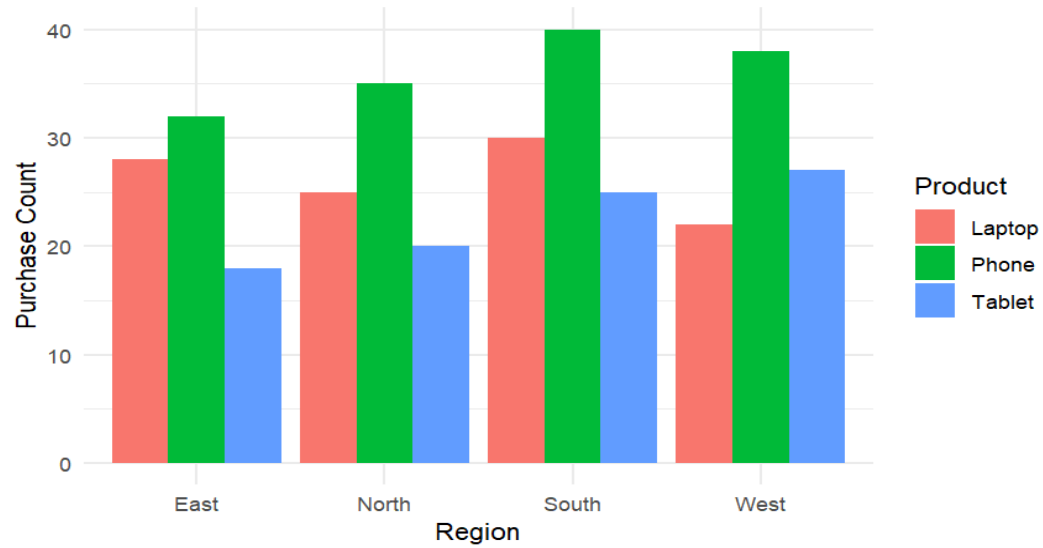
So: MachineLearning has the highest average performance.

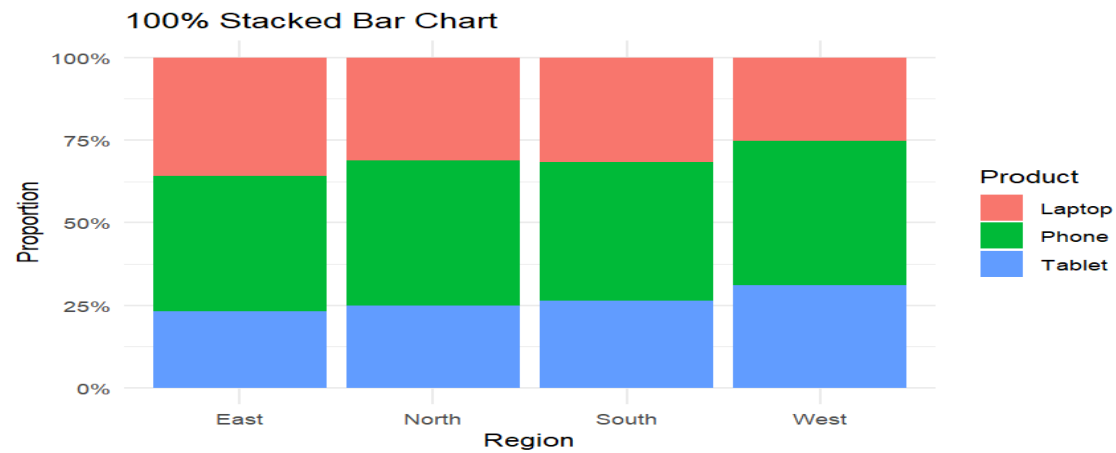
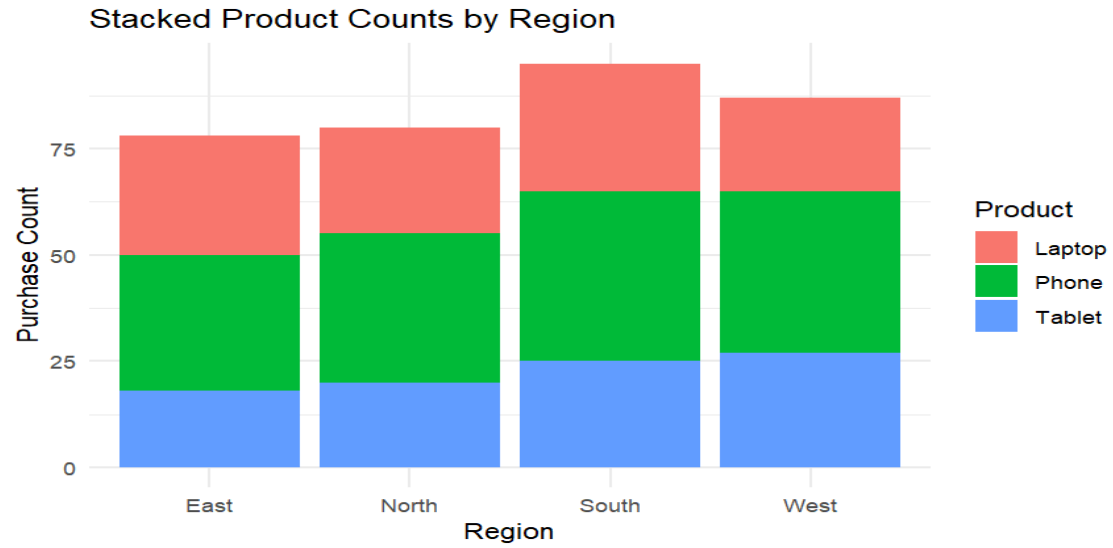
## QUESTION 3 – Pie, Side-by-Side and Stacked Bar Charts

Overall Proportion of Purchases by Product



Product Counts Across Regions





CODE:

```
library(ggplot2)
```

```
library(dplyr)
```

```
# Create dataset
```

```
purchase <- data.frame(
```

```
  Customer_ID = paste0("C", sprintf("%03d", 1:12)),
```

```
  Region = c(
```

```
    "North", "North", "North",
```

```
    "South", "South", "South",
```

```

"East", "East", "East",
"West", "West", "West"),
Product = c(
  "Laptop", "Phone", "Tablet",
  "Laptop", "Phone", "Tablet",
  "Laptop", "Phone", "Tablet",
  "Laptop", "Phone", "Tablet"),
Purchase_Type = c(
  "Online", "Online", "Store",
  "Store", "Online", "Store",
  "Online", "Store", "Online",
  "Store", "Online", "Store"),
Count = c(
  25, 35, 20,
  30, 40, 25,
  28, 32, 18,
  22, 38, 27))
print(purchase)
product_total <- purchase %>%
  group_by(Product) %>%
  summarise(Total = sum(Count))
ggplot(product_total, aes(x = "", y = Total, fill = Product)) +
  geom_bar(stat = "identity", width = 1) +

```

```

coord_polar("y") +

labs(

  title = "Overall Proportion of Purchases by Product" ) +

theme_void()

ggplot(purchase, aes(x = Region, y = Count, fill = Product)) +

geom_bar(

  stat = "identity",

  position = "dodge" ) +

labs(

  title = "Product Counts Across Regions",

  x = "Region",

  y = "Purchase Count" ) +

theme_minimal()

ggplot(purchase, aes(x = Region, y = Count, fill = Product)) +

geom_bar(

  stat = "identity",

  position = "stack" ) +

labs(

  title = "Stacked Product Counts by Region",

  x = "Region",

  y = "Purchase Count" ) +

theme_minimal()

ggplot(purchase, aes(x = Region, y = Count, fill = Product)) +

```

```

geom_bar(

  stat = "identity",

  position = "fill"

) +

labs(

  title = "100% Stacked Bar Chart",

  x = "Region",

  y = "Proportion"

) +

scale_y_continuous(labels = scales::percent) +

theme_minimal()

```

## Interpretation

The **pie chart** is useful for showing the overall proportion of products.

The **side-by-side bar chart** is better for comparing actual product counts between regions.

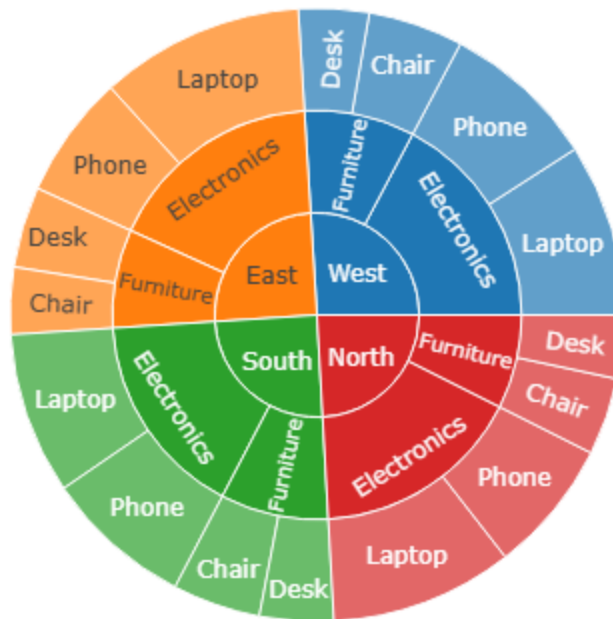
The **stacked bar chart** shows the total purchases as well as product composition.

The **100% stacked bar chart** is the best choice when the objective is to compare **regional proportions**, because every region has the same total height of 100%.

## Important observation

For regional proportions: 100% stacked bar chart is more appropriate than a pie chart because it makes it easier to compare the product composition of different regions.

## QUESTION 4 – Treemap and Sunburst



CODE:

```
sales <- data.frame(
```

```
  Region = c(
    "North","North","North","North",
    "South","South","South","South",
    "East","East","East","East",
    "West","West","West","West"
  ),
```

```
  Category = c(
    "Electronics","Electronics","Furniture","Furniture",
    "Electronics","Electronics","Furniture","Furniture",
    "Electronics","Electronics","Furniture","Furniture",
    "Electronics","Electronics","Furniture","Furniture"
  ),
```

```

Subcategory = c(
  "Laptop","Phone","Chair","Desk",
  "Laptop","Phone","Chair","Desk",
  "Laptop","Phone","Chair","Desk",
  "Laptop","Phone","Chair","Desk"
),

Sales = c(
  52000,38000,22000,18000,
  46000,42000,25000,21000,
  58000,35000,19000,23000,
  49000,44000,27000,20000
)
)

print(sales)

ggplot(
  sales,
  aes(
    area = Sales,
    fill = Category,
    label = Subcategory,
    subgroup = Region,
    subgroup2 = Category
  )
) +

  geom_treemap() +

  geom_treemap_subgroup_border(
    colour = "black",
    size = 2
  ) +

  geom_treemap_subgroup2_border(
    colour = "white",
    size = 1
  ) +

```

```

geom_treemap_text(
  aes(label = paste(Subcategory, "\n$", Sales)),
  colour = "black",
  place = "centre",
  grow = TRUE
) +

labs(
  title = "Company Sales Treemap",
  subtitle = "Region → Category → Subcategory",
  fill = "Category"
) +

theme_minimal()

```

# Region totals

```

region <- aggregate(
  Sales ~ Region,
  data = sales,
  sum
)

```

# Region + Category totals

```

category <- aggregate(
  Sales ~ Region + Category,
  data = sales,
  sum
)

```

# Create Region IDs

```

region$id <- region$Region

```

```

region$parent <- ""

```

```
# Create Category IDs
```

```
category$id <- paste(  
  category$Region,  
  category$Category,  
  sep = "-"  
)
```

```
category$parent <- category$Region
```

```
# Create Subcategory IDs
```

```
subcategory <- sales
```

```
subcategory$id <- paste(  
  subcategory$Region,  
  subcategory$Category,  
  subcategory$Subcategory,  
  sep = "-"  
)
```

```
subcategory$parent <- paste(  
  subcategory$Region,  
  subcategory$Category,  
  sep = "-"  
)
```

```
# Combine all three levels
```

```
sunburst_data <- rbind(  
  
  data.frame(  
    id = region$id,  
    label = region$Region,  
    parent = region$parent,  
    value = region$Sales
```

```

),

data.frame(
  id = category$id,
  label = category$Category,
  parent = category$parent,
  value = category$Sales
),

data.frame(
  id = subcategory$id,
  label = subcategory$Subcategory,
  parent = subcategory$parent,
  value = subcategory$Sales
)
)

```

```

# Create Sunburst

```

```

plot_ly(
  data = sunburst_data,
  ids = ~id,
  labels = ~label,
  parents = ~parent,
  values = ~value,
  type = "sunburst",
  branchvalues = "total"
)

```

```

# Largest Region

```

```

region_sales <- aggregate(
  Sales ~ Region,
  data = sales,
  sum
)

```

```
print(region_sales)
```

```
largest_region <- region_sales[  
  which.max(region_sales$Sales),  
]
```

```
print(largest_region)
```

```
# Largest Category
```

```
category_sales <- aggregate(  
  Sales ~ Category,  
  data = sales,  
  sum  
)
```

```
print(category_sales)
```

```
largest_category <- category_sales[  
  which.max(category_sales$Sales),  
]
```

```
print(largest_category)
```

```
# Largest Subcategory
```

```
subcategory_sales <- aggregate(  
  Sales ~ Subcategory,  
  data = sales,  
  sum  
)
```

```
print(subcategory_sales)
```

```
largest_subcategory <- subcategory_sales[  
  which.max(subcategory_sales$Sales),  
]
```

```
print(largest_subcategory)
```

## Interpretation

- **West is the largest region**, with total sales of **140,000**.
- **Electronics is the largest category**, with total sales of **364,000**, which is higher than Furniture.
- **Laptop is the largest subcategory**, with total sales of **205,000**.
- The **treemap** is easier for comparing sales sizes, while the **sunburst chart** is better for showing the hierarchy of *Region → Category → Subcategory*.
- A limitation of the **treemap** is that small areas can be difficult to read, while the **sunburst** can become crowded when there are many categories.

## Highest-Selling Region

### West

Average/Total Sales:

North = 130,000

South = 134,000

East = 135,000

West = 140,000

So: **West has the highest total sales.**

## Highest-Selling Category

### Electronics

Total Sales:

Electronics = 364,000

Furniture = 175,000

So: **Electronics has the highest total sales.**

## Highest-Selling Subcategory

### Laptop

Total Sales:

Laptop = 205,000

Phone = 159,000

Chair = 93,000

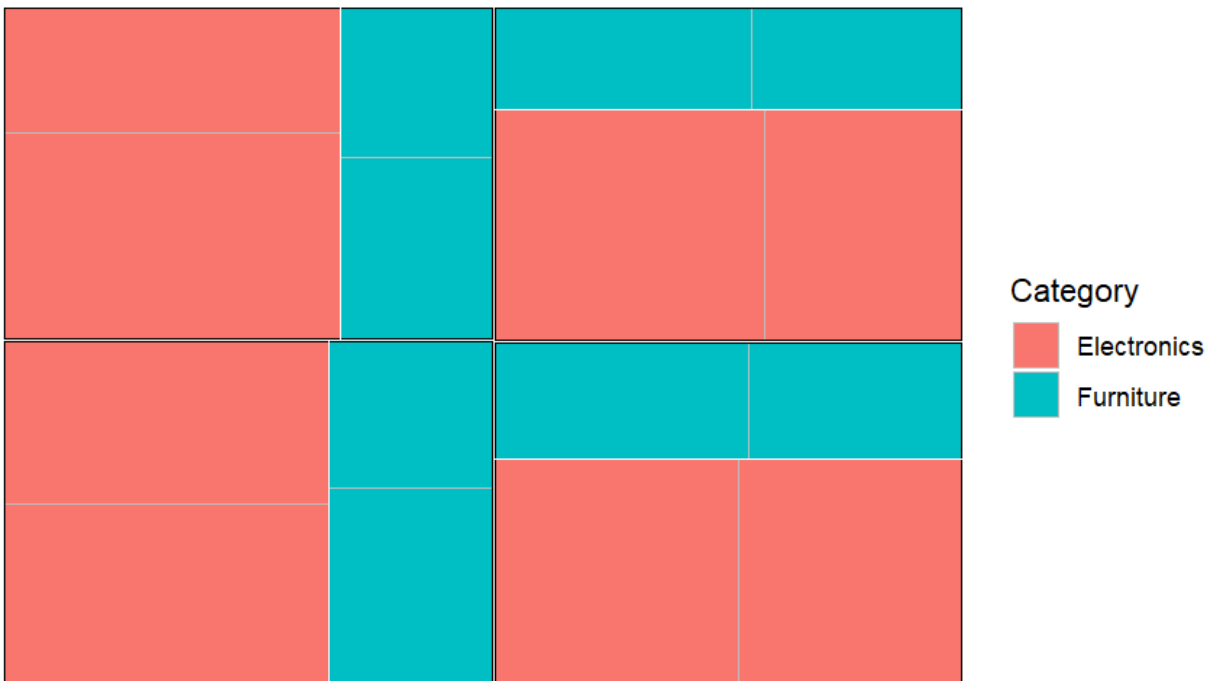
Desk = 82,000

**So: Laptop has the highest total sales**

## Question 5 – Flow-Based Proportions with Sankey Diagram and Parallel Sets

### Company Sales Treemap

Region → Category → Subcategory



#### CODE:

```
install.packages("plotly")
install.packages("ggplot2")
install.packages("ggalluvial")
```

```
library(plotly)
library(ggplot2)
library(ggalluvial)
```

```
career <- data.frame(
```

```
  Student_Group = c(
    "G1", "G2", "G3",
```

```
"G4","G5","G6",  
"G7","G8","G9",  
"G10","G11","G12"  
)
```

```
Specialization = c(  
  "Data Analytics",  
  "Data Analytics",  
  "Data Analytics",  
  "AI",  
  "AI",  
  "AI",  
  "Cyber Security",  
  "Cyber Security",  
  "Cyber Security",  
  "Cloud Computing",  
  "Cloud Computing",  
  "Cloud Computing"  
)
```

```
Internship = c(  
  "Yes","Yes","No",  
  "Yes","Yes","No",  
  "Yes","Yes","No",  
  "Yes","Yes","No"  
)
```

```
Placement_Sector = c(  
  "IT",  
  "Finance",  
  "IT",  
  "IT",  
  "Healthcare",  
  "IT",  
  "IT",  
  "Finance",  
  "Government",  
  "IT",  
  "Finance",  
  "Government"
```

```

),

Students = c(
  18,12,8,
  20,10,6,
  16,9,7,
  15,8,5
)
)

print(career)

# Create nodes

nodes <- data.frame(
  name = c(
    "Data Analytics",
    "AI",
    "Cyber Security",
    "Cloud Computing",
    "Yes",
    "No",
    "IT",
    "Finance",
    "Healthcare",
    "Government"
  )
)

# Find source and target positions

source1 <- match(
  career$Specialization,
  nodes$name
) - 1

target1 <- match(
  career$Internship,

```

```
nodes$name  
) - 1
```

```
source2 <- match(  
  career$Internship,  
  nodes$name  
) - 1
```

```
target2 <- match(  
  career$Placement_Sector,  
  nodes$name  
) - 1
```

```
# Create links
```

```
links <- data.frame(  
  source = c(source1, source2),  
  target = c(target1, target2),  
  value = c(career$Students, career$Students)  
)
```

```
# Create Sankey diagram
```

```
fig <- plot_ly(  
  type = "sankey",  
  orientation = "h",
```

```
  node = list(  
    label = nodes$name,  
    pad = 15,  
    thickness = 20  
  ),
```

```
  link = list(  
    source = links$source,  
    target = links$target,  
    value = links$value  
  )
```

```
)
```

```
fig <- fig %>%
```

```
  layout(
```

```
    title = "Student Career Path Sankey Diagram"
```

```
  )
```

```
fig
```

```
ggplot(
```

```
  career,
```

```
  aes(
```

```
    axis1 = Specialization,
```

```
    axis2 = Internship,
```

```
    axis3 = Placement_Sector,
```

```
    y = Students
```

```
  )
```

```
) +
```

```
  geom_alluvium(
```

```
    aes(fill = Specialization),
```

```
    alpha = 0.7
```

```
  ) +
```

```
  geom_stratum() +
```

```
  geom_text(
```

```
    stat = "stratum",
```

```
    aes(label = after_stat(stratum))
```

```
  ) +
```

```
  scale_x_discrete(
```

```
    limits = c(
```

```
      "Specialization",
```

```
      "Internship",
```

```
      "Placement Sector"
```

```
    )
```

```
  ) +
```

```
  labs(
```

```
title = "Parallel Sets of Student Career Paths",  
x = "Career Path",  
y = "Number of Students",  
fill = "Specialization"  
) +
```

```
theme_minimal()
```

```
internship_total <- aggregate(  
  Students ~ Internship,  
  data = career,  
  sum  
)
```

```
print(internship_total)
```

```
placement_total <- aggregate(  
  Students ~ Placement_Sector,  
  data = career,  
  sum  
)
```

```
print(placement_total)
```

```
specialization_total <- aggregate(  
  Students ~ Specialization,  
  data = career,  
  sum  
)
```

```
print(specialization_total)
```

## **Interpretation**

1. **AI → Yes → IT is the largest individual flow with 20 students.**
2. **Students with internships (108) are much higher than students without internships (26).**
3. **IT is the most common placement sector with 83 students.**
4. **The Sankey diagram communicates flow structure more clearly because wider bands represent more students.**
5. **Incorrect scales, colors, or labels can make a small flow look larger than it actually is and lead to a misleading conclusion.**

## **Internship**

**Yes = 108**

**No = 26**

## **Placement Sector**

**IT = 83**

**Finance = 39**

**Healthcare = 10**

**Government = 12**

## **Specialization**

**Data Analytics = 38**

**AI = 36**

**Cyber Security = 32**

**Cloud Computing = 28**

## Major Flow

The largest individual flow is:

AI → Yes → IT = 20 students

Other major flows:

- Data Analytics → Yes → IT = 18
- Cyber Security → Yes → IT = 16
- Cloud Computing → Yes → IT = 15